

Characterizing Genetic Diversity

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Characterizing Genetic Diversity of Black Bear Populations in Georgia

In this markdown, we'll calculate some simple summaries of genetic diversity for each of the populations represented in our dataset: i) central GA historic (~2012), ii) central GA current (2022-2024), iii) north GA, and iv) south GA. We will calculate the following diversity summaries for each of these:

1. Number of polymorphic SNPs in the population (K)
2. Expected heterozygosity (H_e)
3. Observed heterozygosity (H_o)
4. Inbreeding coefficient (F_{IS})
5. Rarefied allelic richness (AR)

Extract SNP data from filtered VCF

First, use `vcftools` to generate a 012 format file. This file stores genotypes at all SNPs as counts of non-reference allele copies (*i.e.*, the ALT allele) - 0, 1, or 2. This line will generate 3 files:

1. `PopGen.012` - the genotype matrix, individuals are rows, loci are columns
2. `PopGen.012.indv` - a vector of individual IDs (the `rgID`, in this case)
3. `PopGen.012.pos` - a two-column file with `CHROM` and `POS` for each SNP in the dataset

```
ulimit -n 3000
vcftools --vcf ../snppfiltering/PopGen.vcf --out PopGen --012
```

VCftools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snppfiltering/PopGen.vcf
--012
--out PopGen
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
After filtering, kept 260 out of 260 Individuals
Writing 012 matrix files ... Done.
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 1.00 seconds
```

Before we start calculating statistics for the populations, we need to create a file that will facilitate splitting up the dataset. Start with PopGen.012.indv and add columns for the UGA ID number and the region (population) where the sample comes from.

```
#Load metadata file
samps <- read.csv("../input/sampleMetaData_cloneID.csv", header = TRUE)

#Build a popmap file to map individual samples back to their population
popmap <- read.table("PopGen.012.indv", header = FALSE)
names(popmap) <- "vcfID"

popmap$ugaID <- NA
popmap$region <- popmap$year <- NA
popmap$survivor <- popmap$finalClone <- NA
for(i in 1:nrow(popmap)) {
  popmap$region[i] <- samps$region[which(samps$vcfID == popmap$vcfID[i])]
  popmap$ugaID[i] <- samps$ugaID[which(samps$vcfID == popmap$vcfID[i])]
  popmap$year[i] <- samps$year[which(samps$vcfID == popmap$vcfID[i])]
  popmap$finalClone[i] <- samps$finalClone[which(samps$vcfID == popmap$vcfID[i])]
}
```

```

#How many regions for this clone?
nReg <- length(unique(samps$region[which(samps$finalClone ==
                                         popmap$finalClone[i])]))

if(nReg > 1) {
  popmap$survivor[i] <- TRUE
} else {
  popmap$survivor[i] <- FALSE
}
}
nrow(popmap)

```

```
## [1] 260
```

```
head(popmap)
```

```
##           vcfID  ugaID year region finalClone survivor
## 1 UGA_173101_P020_WC08 336-23 2023   CGA         3     TRUE
## 2 UGA_173101_P020_WC01 359-23 2023   CGA         5    FALSE
## 3 UGA_173101_P020_WB12 110-23 2023   CGA         6    FALSE
## 4 UGA_173101_P020_WB10  79-24 2024   CGA         7    FALSE
## 5 UGA_173101_P020_WB05 492-24 2024   CGA        340    FALSE
## 6 UGA_173101_P019_WH08 80-NGA 2024   NGA         18    FALSE
```

```
table(popmap$region)
```

```
##
##   CGA CGA-H  NGA  SGA
##   185   16   31   28
```

```
table(popmap$year[popmap$region == "CGA"])
```

```
##
## 2022 2023 2024
##    3  101  81
```

```
table(popmap$year[popmap$region == "CGA-H"])
```

```
##
## 2012
##    16
```

```
sum(popmap$survivor)
```

```
## [1] 3
```

Now read in the 012 format SNP data...

```
snps <- read.table("PopGen.012", header = FALSE)
dim(snps) #rows are individuals, columns are SNPs
```

```
## [1] 260 10718
```

```
dim(popmap)
```

```
## [1] 260 6
```

```
#Missing data are coded as -1's
```

```
#Drop all samples with fewer than 2000 genotyped loci
```

```
droprows <- which(rowSums(snps[, -1] > -1) < 2000)
```

```
snps <- snps[-droprows,]
```

```
dim(snps)
```

```
## [1] 238 10718
```

```
popmap <- popmap[-droprows,]
```

```
dim(popmap)
```

```
## [1] 238 6
```

```
table(popmap$region)
```

```
##
```

```
## CGA CGA-H NGA SGA
```

```
## 168 13 31 26
```

```
table(popmap$year[popmap$region == "CGA"])
```

```
##
```

```
## 2022 2023 2024
```

```
## 3 90 75
```

```
table(popmap$year[popmap$region == "CGA-H"])
```

```
##
```

```
## 2012
```

```
## 13
```

Create subset datasets for each “population” in our dataset. Don’t forget to include bears captured in both 2012 and 2023 (there are 2 of these) in *both* CGA23 and CGA12! We will create four of these objects:

1. The focal population in central GA sampled in this project - CGA23
2. Historic samples from central GA (from ~2012) - CGA12
3. The north GA population - NGA
4. The south GA population - SGA

```
#Sample sizes per "region"  
table(popmap$region)
```

```
##  
##   CGA CGA-H   NGA   SGA  
##   168   13    31    26
```

```
# but there are some survivors too...  
popmap$year[which(popmap$survivor == TRUE)]
```

```
## [1] 2023 2023 2012
```

```
# these bears should be represented in both CGA12 and CGA23
```

```
#Make CGA12 & CGA23 snp datasets  
CGA12snps <- snps[which(popmap$region == "CGA-H" | popmap$survivor == TRUE),]  
CGA12pop <- popmap[which(popmap$region == "CGA-H" | popmap$survivor == TRUE),]  
CGA23snps <- snps[which(popmap$region == "CGA" | popmap$survivor == TRUE),]  
CGA23pop <- popmap[which(popmap$region == "CGA" | popmap$survivor == TRUE),]  
nrow(CGA12snps)
```

```
## [1] 15
```

```
nrow(CGA12pop)
```

```
## [1] 15
```

```
nrow(CGA23snps)
```

```
## [1] 169
```

```
nrow(CGA23pop)
```

```
## [1] 169
```

```
NGAsnps <- snps[popmap$region == "NGA",]  
NGApop <- popmap[popmap$region == "NGA",]  
nrow(NGAsnps)
```

```
## [1] 31
```

```
nrow(NGApop)
```

```
## [1] 31
```

```
SGAsnps <- snps[popmap$region == "SGA",]
SGApop <- popmap[popmap$region == "SGA",]
nrow(SGAsnps)
```

```
## [1] 26
```

```
nrow(SGApop)
```

```
## [1] 26
```

Number of polymorphic markers per population

Record the number of SNPs that are variable within each of the four populations...

```
#Function to check polymorphism
is.poly <- function(x) {
  if(any(x == 1)) { #If any individual is a heterozygote
    out <- TRUE
  } else if(any(x == 0) & any(x == 2)) { #If both homozygotes are observed
    out <- TRUE
  } else { #If no hets and both homozygotes NOT observed, not polymorphic
    out <- FALSE
  }
  return(out)
}

#Polymorphic SNP numbers per population
sum(apply(snps[,-1], 2, FUN=is.poly))
```

```
## [1] 10714
```

```
sum(apply(CGA12snps[,-1], 2, FUN=is.poly))
```

```
## [1] 3283
```

```
sum(apply(CGA23snps[,-1], 2, FUN=is.poly))
```

```
## [1] 9440
```

```
sum(apply(NGAsnps[,-1], 2, FUN=is.poly))
```

```
## [1] 5503
```

```
sum(apply(SGAsnps[,-1], 2, FUN=is.poly))
```

```
## [1] 6120
```

Expected Heterozygosity

Calculate allele frequencies per locus, then expected heterozygosity of that locus...

```
getHe <- function(x) {  
  frq <- sum(x[x > -1])/(2*(sum(x > -1)))  
  He.loc <- 2*frq*(1-frq)  
  return(He.loc)  
}  
  
mean(apply(CGA12snps[, -1], 2, FUN=getHe), na.rm = TRUE)
```

```
## [1] 0.07450451
```

```
mean(apply(CGA23snps[, -1], 2, FUN=getHe), na.rm = TRUE)
```

```
## [1] 0.07790806
```

```
mean(apply(NGAsnps[, -1], 2, FUN=getHe), na.rm = TRUE)
```

```
## [1] 0.110205
```

```
mean(apply(SGAsnps[, -1], 2, FUN=getHe), na.rm = TRUE)
```

```
## [1] 0.1124583
```

Observed Heterozygosity

Calculate observed heterozygosity (# hets / # genotyped) per locus, then average across loci...

```
#Calculating per locus, then averaging across loci  
getHo <- function(x) {  
  n.hets <- sum(x == 1)  
  n.geno <- sum(x > -1)  
  Ho.loc <- n.hets/n.geno  
  return(Ho.loc)  
}  
  
mean(apply(CGA12snps[, -1], 2, FUN=getHo), na.rm = TRUE)
```

```
## [1] 0.07597949
```

```
mean(apply(CGA23snps[, -1], 2, FUN=getHo), na.rm = TRUE)
```

```
## [1] 0.07734437
```

```
mean(apply(NGAsnps[, -1], 2, FUN=getHo), na.rm = TRUE)
```

```
## [1] 0.1090443
```

```
mean(apply(SGAsnps[, -1], 2, FUN=getHo), na.rm = TRUE)
```

```
## [1] 0.1138383
```

Inbreeding Coefficients

Now use `vcftools` to calculate inbreeding coefficients - F_{IS} - for individuals, average them for each of the populations, and look to see if there's evidence for highly inbred individuals.

First, make keep files for the population to use with `vcftools`.

```
#Make keep files for the individual populations - with these same individuals...
write.table(CGA12pop$vcfID, file = "CGA12keep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
write.table(CGA23pop$vcfID, file = "CGA23keep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
write.table(NGApop$vcfID, file = "NGAkeep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
write.table(SGApop$vcfID, file = "SGAkeep.txt",
            quote = FALSE,
            row.names = FALSE,
            col.names = FALSE)
```

Now use `vcftools` to calculate F_{IS} .

```
vcftools --vcf ../snppfiltering/PopGen.vcf \
         --keep CGA12keep.txt --out CGA12 --het

vcftools --vcf ../snppfiltering/PopGen.vcf \
         --keep CGA23keep.txt --out CGA23 --het

vcftools --vcf ../snppfiltering/PopGen.vcf \
         --keep NGAkeep.txt --out NGA --het

vcftools --vcf ../snppfiltering/PopGen.vcf \
         --keep SGAkeep.txt --out SGA --het
```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snpfiltering/PopGen.vcf
--keep CGA12keep.txt
--het
--out CGA12
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 15 out of 260 Individuals
Outputting Individual Heterozygosity
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 0.00 seconds
```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snpfiltering/PopGen.vcf
--keep CGA23keep.txt
--het
--out CGA23
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placeme
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
```

Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
 Keeping individuals in 'keep' list
 After filtering, kept 169 out of 260 Individuals
 Outputting Individual Heterozygosity
 After filtering, kept 10717 out of a possible 10717 Sites
 Run Time = 1.00 seconds

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snppfiltering/PopGen.vcf
--keep NGAkeep.txt
--het
--out NGA
```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
 Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
 Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
 Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
 Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
 Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
 Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
 Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
 Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
 Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
 Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
 Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
 Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
 Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
 Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
 Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
 Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
 Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
 Keeping individuals in 'keep' list
 After filtering, kept 31 out of 260 Individuals
 Outputting Individual Heterozygosity
 After filtering, kept 10717 out of a possible 10717 Sites
 Run Time = 0.00 seconds

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snppfiltering/PopGen.vcf
--keep SGAkeep.txt
--het
--out SGA
```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
 Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
 Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
 Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
 Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
 Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance

```

Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 26 out of 260 Individuals
Outputting Individual Heterozygosity
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 0.00 seconds

```

Read in the output, calculate means, ranges, and the number of individuals with $F_{IS} > 0.15$...

```

CGA12.Fis <- read.table("CGA12.het", header = TRUE)
mean(CGA12.Fis$F)

```

```
## [1] 0.03863467
```

```
range(CGA12.Fis$F)
```

```
## [1] -0.35598 0.44566
```

```
sum(CGA12.Fis$F > 0.15)
```

```
## [1] 3
```

```

CGA23.Fis <- read.table("CGA23.het", header = TRUE)
mean(CGA23.Fis$F)

```

```
## [1] 0.01587254
```

```
range(CGA23.Fis$F)
```

```
## [1] -0.52392 0.63136
```

```
sum(CGA23.Fis$F > 0.15)
```

```
## [1] 49
```

```

NGA.Fis <- read.table("NGA.het", header = TRUE)
mean(NGA.Fis$F)

```

```
## [1] 0.01876548
```

```
range(NGA.Fis$F)
```

```
## [1] -0.34485 0.15652
```

```
sum(NGA.Fis$F > 0.15)
```

```
## [1] 1
```

```
SGA.Fis <- read.table("SGA.het", header = TRUE)  
mean(SGA.Fis$F)
```

```
## [1] 0.0048
```

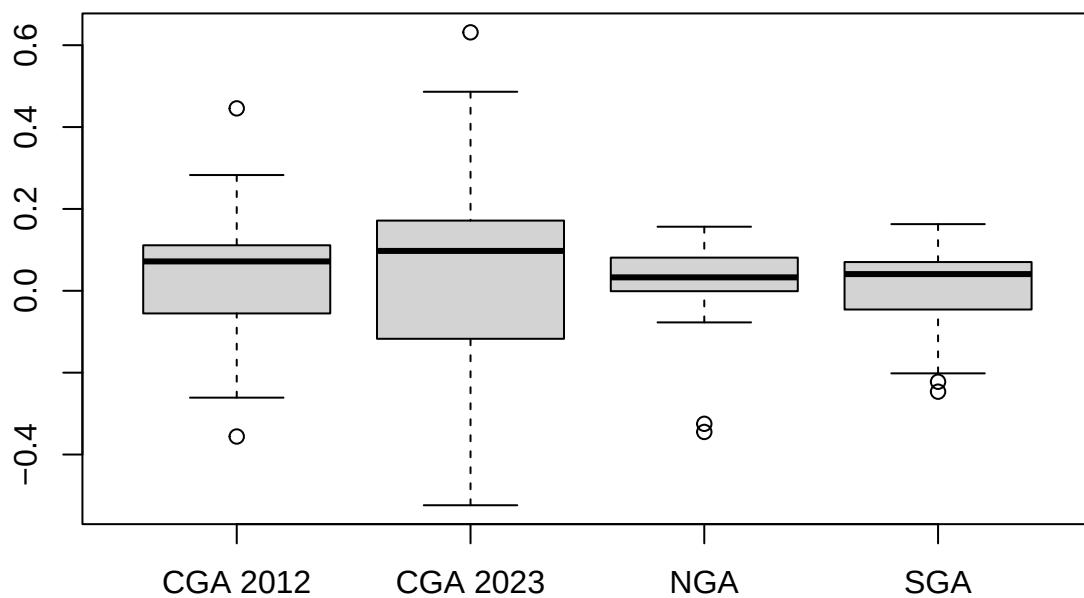
```
range(SGA.Fis$F)
```

```
## [1] -0.24647 0.16285
```

```
sum(SGA.Fis$F > 0.15)
```

```
## [1] 1
```

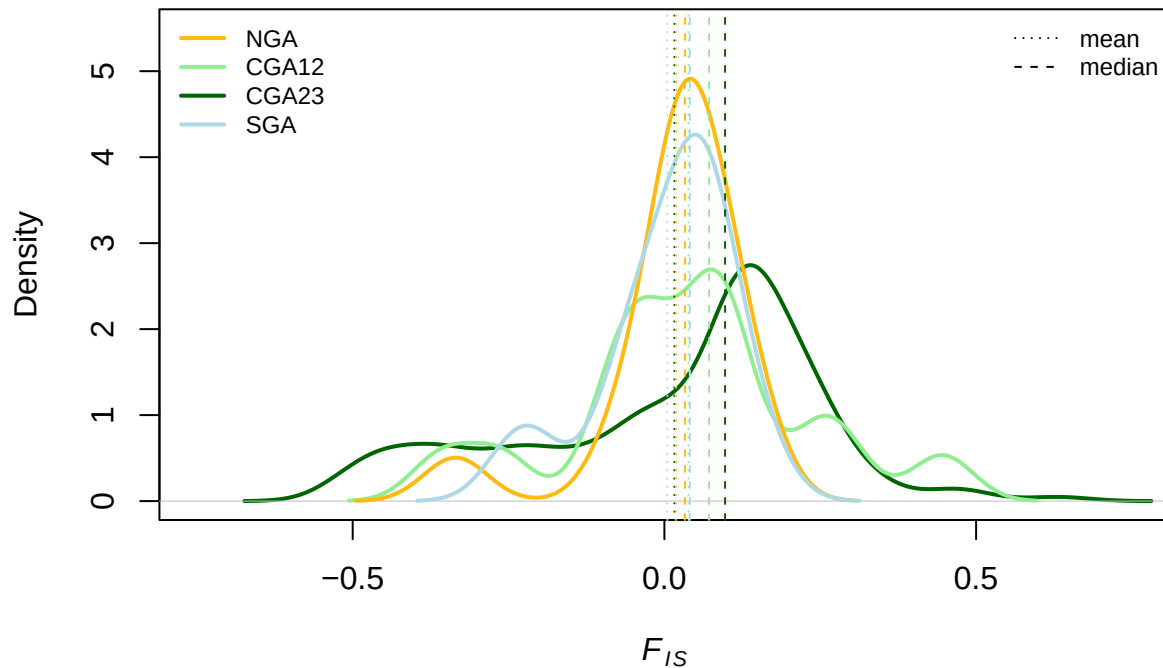
```
#Visualize the distribution of Fis  
boxplot(CGA12.Fis$F, CGA23.Fis$F, NGA.Fis$F, SGA.Fis$F,  
        names = c("CGA 2012", "CGA 2023", "NGA", "SGA"))
```



```

# as density plots?
plot(density(CGA23.Fis$F, bw = 0.05),
     lwd = 2, col = "darkgreen",
     xlim = c(-0.75, 0.75), ylim = c(0, 5.5),
     main = "", xlab = expression(italic(F[IS])))
lines(density(CGA12.Fis$F, bw = 0.05), lwd = 2, col = "lightgreen")
lines(density(NGA.Fis$F, bw = 0.05), lwd = 2, col = "darkgoldenrod1")
lines(density(SGA.Fis$F, bw = 0.05), lwd = 2, col = "lightblue")
abline(v = mean(CGA23.Fis$F), lty = "dotted", col = "darkgreen")
abline(v = mean(CGA12.Fis$F), lty = "dotted", col = "lightgreen")
abline(v = mean(NGA.Fis$F), lty = "dotted", col = "darkgoldenrod1")
abline(v = mean(SGA.Fis$F), lty = "dotted", col = "lightblue")
abline(v = median(CGA23.Fis$F), lty = "dashed", col = "darkgreen")
abline(v = median(CGA12.Fis$F), lty = "dashed", col = "lightgreen")
abline(v = median(NGA.Fis$F), lty = "dashed", col = "darkgoldenrod1")
abline(v = median(SGA.Fis$F), lty = "dashed", col = "lightblue")
legend("topleft", lwd = 2, bty = "n", cex=0.75,
      col = c("darkgoldenrod1","lightgreen","darkgreen","lightblue"),
      legend = c("NGA","CGA12","CGA23","SGA"))
legend("topright", bty = "n", lwd = 1, cex = 0.75,
      lty = c("dotted", "dashed"), legend = c("mean", "median"))

```



Note that, while the mean F_{IS} values are quite similar across populations, the median value in contemporary CGA samples is quite a bit higher ($F_{IS} = 0.09736$).

Allelic Richness

We'll use the `pegas` R package to calculate rarefied allelic richness for each of the populations in our dataset. To do this, we first need to create VCF files for each of the groupings, rename historic samples...

```
vcftools --vcf ../snpfiltering/PopGen.vcf --keep CGA12keep.txt --recode --out CGA12
vcftools --vcf ../snpfiltering/PopGen.vcf --keep CGA23keep.txt --recode --out CGA23
vcftools --vcf ../snpfiltering/PopGen.vcf --keep NGAkeep.txt --recode --out NGA
vcftools --vcf ../snpfiltering/PopGen.vcf --keep SGAkeep.txt --recode --out SGA

#Change names for CGA12 to avoid overlap with CGA23 survivors
sed -e 's/UGA/UGAh/g' CGA12.recode.vcf > CGA12h.recode.vcf

#Grab header and SNP rows
grep -v ^## CGA12h.recode.vcf > CGA12snps
grep -v ^## CGA23.recode.vcf | cut -f 10- > CGA23snps
grep -v ^## NGA.recode.vcf | cut -f 10- > NGAsnps
grep -v ^## SGA.recode.vcf | cut -f 10- > SGAsnps

#Paste data columns together
paste CGA12snps CGA23snps NGAsnps SGAsnps > tempcombined

#Grab header from one of the files
grep ^## CGA12.recode.vcf > combined.vcf
cat tempcombined >> combined.vcf

#Make an individual ID file
grep -v ^## combined.vcf | head -1 | tr '\t' '\n' | grep ^UGA > IID.txt
```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```
--vcf ../snpfiltering/PopGen.vcf
--keep CGA12keep.txt
--out CGA12
--recode
```

```
Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
```

```

Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 15 out of 260 Individuals
Outputting VCF file...
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 1.00 seconds

```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```

--vcf ../snpfiltering/PopGen.vcf
--keep CGA23keep.txt
--out CGA23
--recode

```

```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 169 out of 260 Individuals
Outputting VCF file...
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 0.00 seconds

```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```

--vcf ../snpfiltering/PopGen.vcf
--keep NGAkeep.txt
--out NGA
--recode

```

```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance

```

```

Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 31 out of 260 Individuals
Outputting VCF file...
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 1.00 seconds

```

VCFTools - 0.1.17

(C) Adam Auton and Anthony Marcketta 2009

Parameters as interpreted:

```

--vcf ../snppfiltering/PopGen.vcf
--keep SGAkeep.txt
--out SGA
--recode

```

```

Warning: Expected at least 2 parts in INFO entry: ID=AF,Number=A,Type=Float,Description="Estimated alle
Warning: Expected at least 2 parts in INFO entry: ID=PRO,Number=1,Type=Float,Description="Reference alle
Warning: Expected at least 2 parts in INFO entry: ID=PAO,Number=A,Type=Float,Description="Alternate alle
Warning: Expected at least 2 parts in INFO entry: ID=SRP,Number=1,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=SAP,Number=A,Type=Float,Description="Strand balance
Warning: Expected at least 2 parts in INFO entry: ID=AB,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=ABP,Number=A,Type=Float,Description="Allele balance
Warning: Expected at least 2 parts in INFO entry: ID=RPP,Number=A,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=RPPR,Number=1,Type=Float,Description="Read Placemen
Warning: Expected at least 2 parts in INFO entry: ID=EPP,Number=A,Type=Float,Description="End Placement
Warning: Expected at least 2 parts in INFO entry: ID=EPPR,Number=1,Type=Float,Description="End Placemen
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=TYPE,Number=A,Type=String,Description="The type of
Warning: Expected at least 2 parts in INFO entry: ID=CIGAR,Number=A,Type=String,Description="The extend
Keeping individuals in 'keep' list
After filtering, kept 26 out of 260 Individuals
Outputting VCF file...
After filtering, kept 10717 out of a possible 10717 Sites
Run Time = 0.00 seconds

```

Now read this combined vcf file into R, create a **genind** object in the R package **adegenet**, remove markers with too few individuals genotyped (below the sample size set for rarefaction), and use the **pegas** package to calculate rarefied allelic richness...


```
library(vcfR)
```

```
##
##      *****      ***   vcfR   ***      *****
##      This is vcfR 1.15.0
##      browseVignettes('vcfR') # Documentation
##      citation('vcfR') # Citation
##      *****      *****      *****      *****
```

```
library(adegenet)
```

```
## Loading required package: ade4

##
##      /// adegenet 2.1.11 is loaded //////////////////////////////////
##
##      > overview: '?adegenet'
##      > tutorials/doc/questions: 'adegenetWeb()'
##      > bug reports/feature requests: adegenetIssues()
```

```
library(pegas)
```

```
## Loading required package: ape

## Registered S3 method overwritten by 'pegas':
##      method      from
##      print.amova ade4

##
## Attaching package: 'pegas'

## The following object is masked from 'package:ape':
##
##      mst

## The following object is masked from 'package:ade4':
##
##      amova

## The following objects are masked from 'package:vcfR':
##
##      getINFO, write.vcf
```

```
IID <- read.table("IID.txt", header = FALSE)
FID <- c()
for(i in 1:nrow(IID)) {
  if(strsplit(IID[i,1],split="_")[[1]][1] == "UGAh") {
    FID[i] <- samps$region[which(samps$vcfID == gsub("h","",IID[i,1]))]
  } else {
```

```

    FID[i] <- samps$region[which(samps$vcfID == IID[i,1])]
  }

  if(FID[i] %in% c("CGA", "CGA-H")) {
    if(strsplit(IID[i,1],split="_")[[1]][1] == "UGAh") {
      FID[i] <- "CGA12"
    } else {
      FID[i] <- "CGA23"
    }
  }
}
}
table(FID)

```

```

## FID
## CGA12 CGA23   NGA   SGA
##    15   169    31    26

```

```

plinkPops <- data.frame(FID=FID, IID=IID, pop=FID)

vcf <- read.vcfR("combined.vcf", verbose=TRUE)

```

```

## Scanning file to determine attributes.
## File attributes:
##   meta lines: 664
##   header_line: 665
##   variant count: 10717
##   column count: 250
## Meta line 664 read in.
## All meta lines processed.
## gt matrix initialized.
## Character matrix gt created.
##   Character matrix gt rows: 10717
##   Character matrix gt cols: 250
##   skip: 0
##   nrows: 10717
##   row_num: 0
## Processed variant 1000Processed variant 2000Processed variant 3000Processed variant 4000Processed variant 5000
## All variants processed

```

```

my_genind <- vcfR2genind(vcf, pop=plinkPops$pop)

n.matrix <- data.frame(
  CGA12 = colSums(!is.na(my_genind@tab[which(my_genind@pop == "CGA12"),])),
  CGA23 = colSums(!is.na(my_genind@tab[which(my_genind@pop == "CGA23"),])),
  NGA = colSums(!is.na(my_genind@tab[which(my_genind@pop == "NGA"),])),
  SGA = colSums(!is.na(my_genind@tab[which(my_genind@pop == "SGA"),]))
)

#Subset to markers with at least "rare.n.ind" individuals genotyped in all pops
rare.n.ind <- 5
rare.n.allele <- 2*rare.n.ind

```

```

min.per.pop <- apply(n.matrix, 1, min)
keep.loc <- unique(
  sapply(names(min.per.pop[which(min.per.pop >= rare.n.ind)]),
    FUN = function(x) strsplit(x, split = ".", fixed=TRUE)[[1]][1])
)
my_genind2 <- my_genind[loc=keep.loc]
sum(levels(my_genind2@loc.fac) %in% keep.loc)

```

```
## [1] 10707
```

```
length(keep.loc)
```

```
## [1] 10707
```

```
length(my_genind2@loc.fac)/2
```

```
## [1] 10705.5
```

```

AR <- allelicrichness(as.loci(my_genind2),
  method = "rarefaction",
  min.n = rare.n.allele)
rowMeans(AR)

```

```

##      CGA12      CGA23      NGA      SGA
## 1.221170 1.220735 1.307912 1.327759

```

Summary of Results

```
knitr::kable(t(out), digits = c(0,1,4,4,3,2))
```

	n	K	He	Ho	Fis	AR
NGA	31	5503	0.1102	0.1090	0.019	1.31
CGA12	15	3283	0.0745	0.0760	0.039	1.22
CGA23	169	9440	0.0779	0.0773	0.016	1.22
SGA	26	6120	0.1125	0.1138	0.005	1.33